

DIGITAL SIGNAL OSCILLOSCOPE

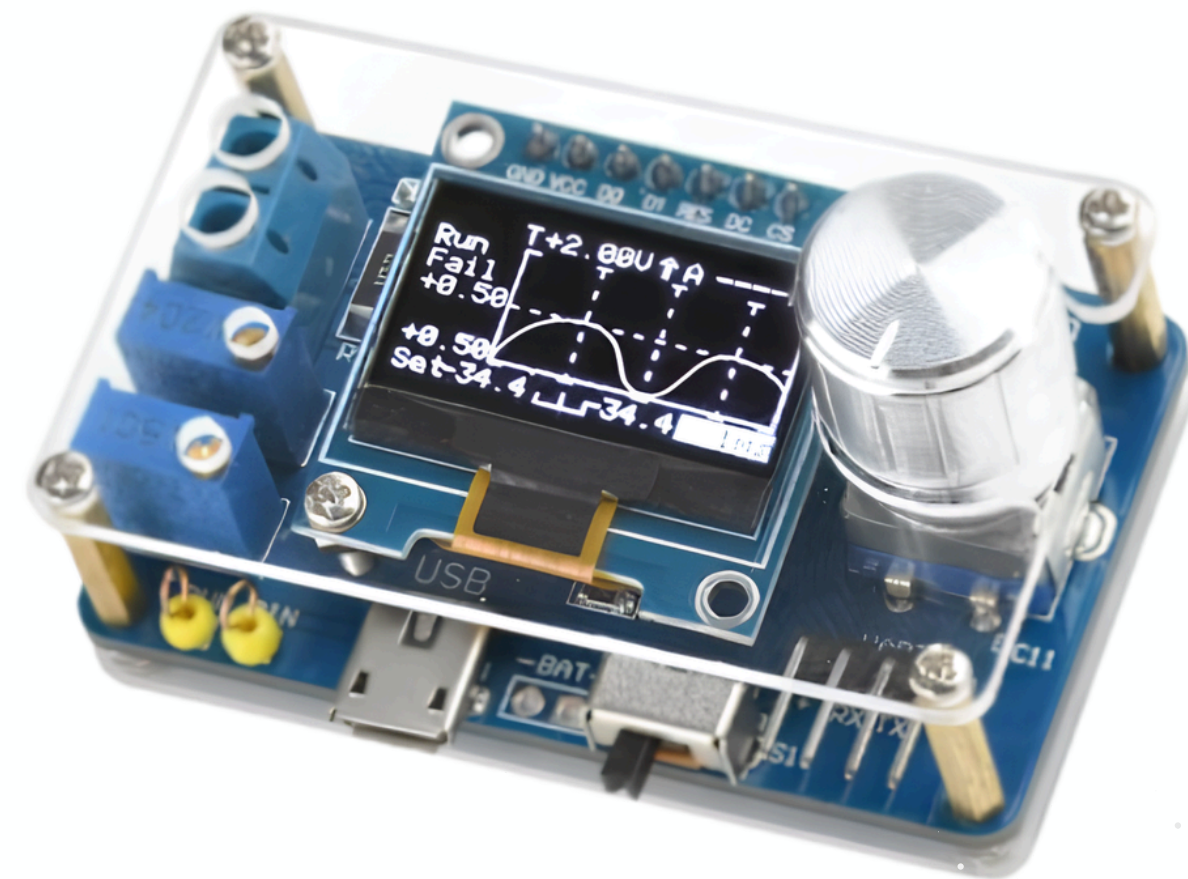
Presented by:

ANURAG KUMAR - 23EC01002

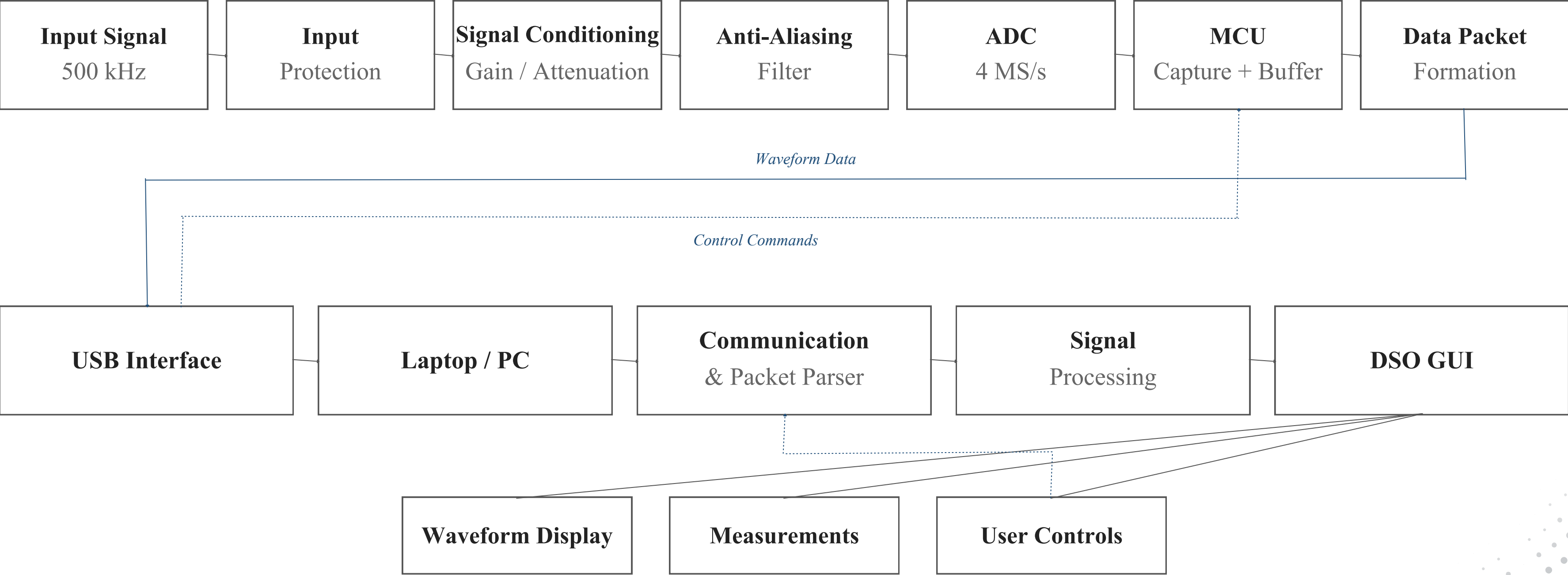
ROHINISH RAJ - 23EC01038

PRASANTH - 23EC01046

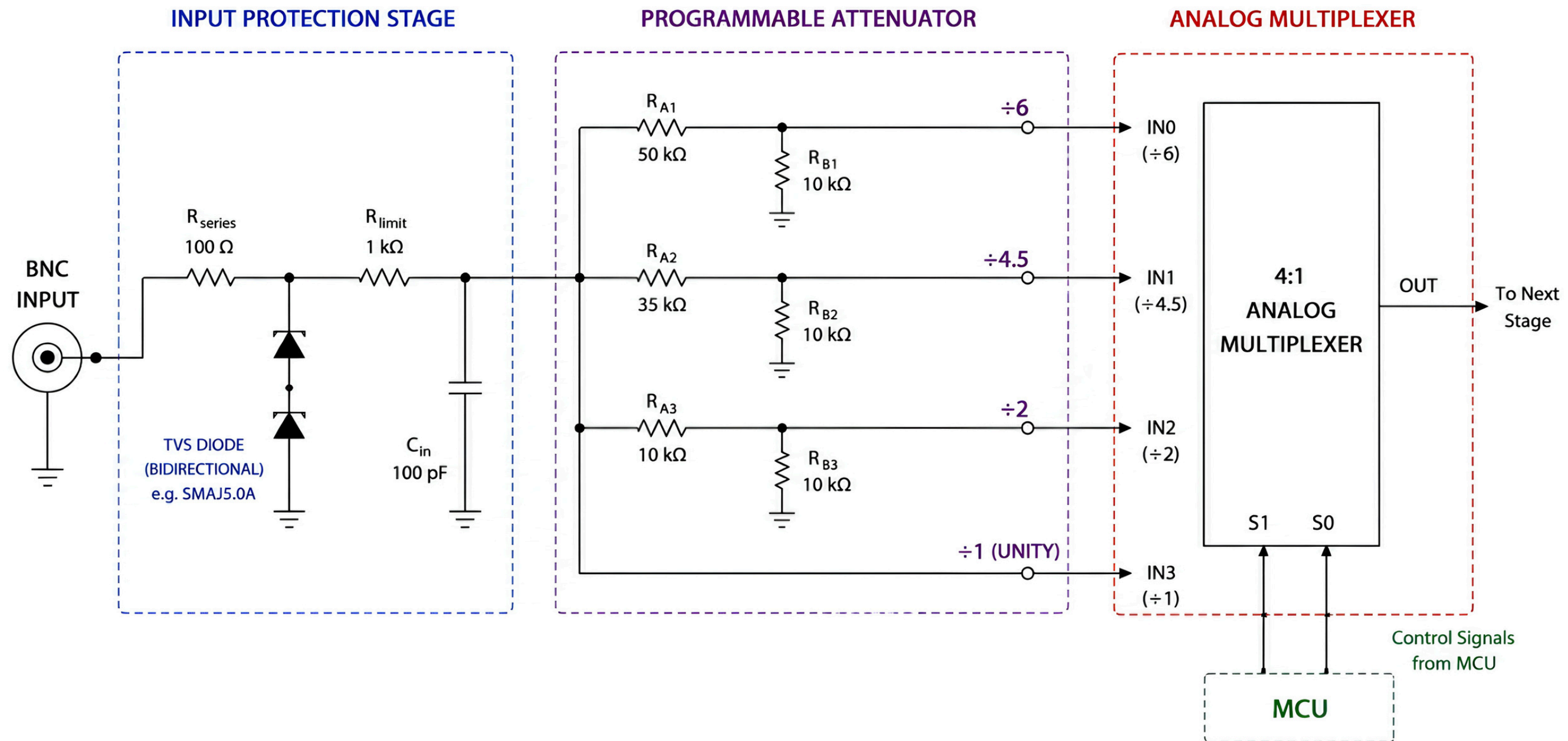
SREENADH - 23EC01052



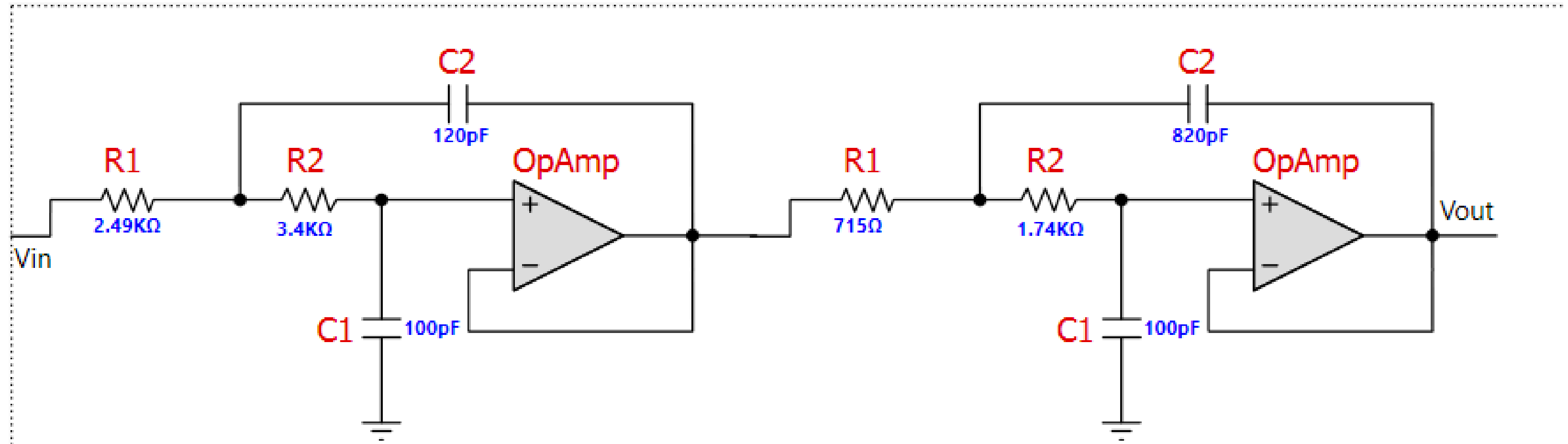
BLOCK DIAGRAM



INPUT PROTECTION AND ATTENUATION



ANTI ALIASING FILTER



ANTI-ALIASING FILTER (AAF) DESIGN

1 Topology

- 4th Order Low-Pass Sallen-Key Filter, Butterworth Response
- 2-Stage Cascaded Active Filter (Ideal Op-Amp based)
- Unity Passband Gain ($A_o = 1 \text{ V/V}$, 0 dB)

2 Design Specifications

- Passband Frequency: 500 kHz
- Corner Frequency Attenuation: -3 dB
- Stopband Frequency: 2 MHz
- Stopband Attenuation: -48 dB
- Allowable Passband Ripple: 1 dB

3 Purpose

- Filters out high-frequency noise/signal components above the Nyquist frequency before ADC sampling, preventing aliasing distortion in the digitized

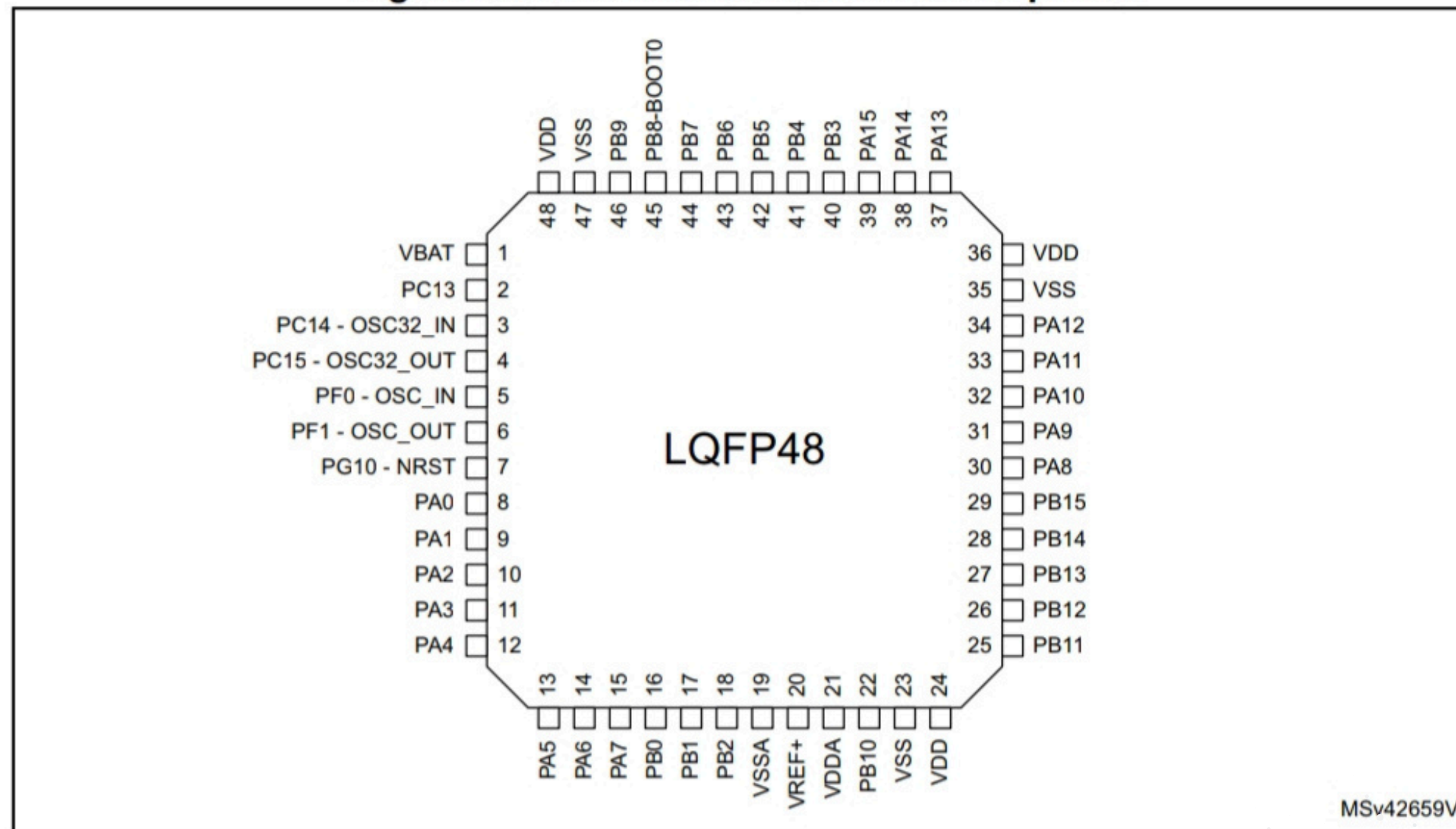
4 Performance

- Flat passband response up to $\sim 500 \text{ kHz}$
- Sharp roll-off ($\sim -48 \text{ dB}$) achieved by 2 MHz, consistent with 4th order Butterworth characteristics

MCU PINOUT

LQFP48 pinout description

Figure 8. STM32G431x6/x8/xB LQFP48 pinout



1. The above figure shows the package top view.

Features

- Core: Arm® 32-bit Cortex®-M4 CPU with FPU, Adaptive real-time accelerator (ART Accelerator) allowing 0-wait-state execution from Flash memory, frequency up to 170 MHz with 213 DMIPS, MPU, DSP instructions
 - Operating conditions:
 - V_{DD} , V_{DDA} voltage range: 1.71 V to 3.6 V
 - Mathematical hardware accelerators
 - CORDIC for trigonometric functions acceleration
 - FMAC: filter mathematical accelerator
 - Memories
 - 128 Kbytes of Flash memory with ECC support, proprietary code readout protection (PCROP), securable memory area, 1 Kbyte OTP
 - 22 Kbytes of SRAM, with hardware parity check implemented on the first 16 Kbytes
 - Routine booster: 10 Kbytes of SRAM on instruction and data bus, with hardware parity check (CCM SRAM)
 - Reset and supply management
 - Power-on/power-down reset (POR/PDR/BOR)
 - Programmable voltage detector (PVD)
 - Low-power modes: sleep, stop, standby and shutdown
 - V_{BAT} supply for RTC and backup registers
 - Clock management
 - 4 to 48 MHz crystal oscillator
 - 32 kHz oscillator with calibration
 - Internal 16 MHz RC with PLL option ($\pm 1\%$)
 - Internal 32 kHz RC oscillator ($\pm 5\%$)
 - Up to 86 fast I/Os
 - All mappable on external interrupt vectors
 - Several I/Os with 5 V tolerant capability
- 

LQFP32 (7 x 7 mm)	UFQFPN32 (5 x 5 mm)	UFBGA64 (5 x 5 mm)	WLCSP49 (Pitch 0.4)
LQFP48 (7 x 7 mm)	UFQFPN48 (7 x 7 mm)		
LQFP64 (10 x 10 mm)			
LQFP80 (12 x 12 mm)			
LQFP100 (14 x 14 mm)			
- Interconnect matrix
 - 12-channel DMA controller
 - 2 x ADCs 0.25 μ s (up to 23 channels). Resolution up to 16-bit with hardware oversampling, 0 to 3.6 V conversion range
 - 4 x 12-bit DAC channels
 - 2 x buffered external channels 1 MSPS
 - 2 x unbuffered internal channels 15 MSPS
 - 4 x ultra-fast rail-to-rail analog comparators
 - 3 x operational amplifiers that can be used in PGA mode, all terminals accessible
 - Internal voltage reference buffer (VREFBUF) supporting three output voltages (2.048 V, 2.5 V, 2.9 V)
 - 14 timers:
 - 1 x 32-bit timer and 2 x 16-bit timers with up to four IC/OC/PWM or pulse counter and quadrature (incremental) encoder input
 - 2 x 16-bit 8-channel advanced motor control timers, with up to 8 x PWM channels, dead time generation and emergency stop
 - 1 x 16-bit timer with 2 x IC/OCs, one OCN/PWM, dead time generation and emergency stop
 - 2 x 16-bit timers with IC/OC/OCN/PWM, dead time generation and emergency stop
 - 2 x watchdog timers (independent, window)
 - 1 x SysTick timer: 24-bit downcounter
 - 2 x 16-bit basic timers
 - 1 x low-power timer



PIN DESCRIPTION

LQFP48 Pin	Pin Name	Target Function
33	PA11	USB_DM
34	PA12	USB_DP
12	PA4	DAC1_OUT1
13	PA5	DAC1_OUT2
9	PA1	OPAMP1_VINP
11	PA3	OPAMP1_VINM
10	PA2	OPAMP1_VOUT
15	PA7	OPAMP2_VINP
13	PA5	OPAMP2_VINM
14	PA6	OPAMP2_VOUT
16	PB0	OPAMP3_VINP
18	PB2	OPAMP3_VINM
17	PB1	OPAMP3_VOUT
8	PA0	ADC12_IN1
25	PB11	ADC12_IN14
26	PB12	ADC1_IN11
28	PB14	ADC1_IN5
29	PB15	ADC2_IN15



MCU requirements					ANALOG frontend				
Component	Recommended Part	Purpose	Qty	COST	Component	Recommended Part	Purpose	Qty	Cost
Microcontroller	STM32G431C8T6 (LQFP-48)	Dual 4-5 MSPS ADCs, DSP, built-in USB PHY	1	461	High-Speed Op-Amp	OPA2350 or AD8032 or NE5532	Used for the Active Anti-Aliasing Filter and input buffering/scaling	1	160
System Crystal (HSE)	8 MHz SMD Crystal (HC-49/US or 3225)	Required for accurate USB timing and high-speed ADC clocking	1	6	Input Connector	PCB-Mount BNC Connector	Standard oscilloscope probe connection	2	252
Load Capacitors	20pF Ceramic (0603)	Paired with the 8 MHz crystal	2	lab	Coupling Cap	1µF Film or Ceramic	Blocks DC voltage when in AC-coupled mode	2	lab
Decoupling Caps	100nF (0.1µF) Ceramic (0603)	Placed as close to every VDD pin as possible to kill digital noise	5	lab	Overvoltage Protection	BAV99 Dual Diode	Clamps voltage spikes so they don't blow up the Op-Amp or ADC	2	18
Bulk Capacitor	4.7µF Ceramic (0805)	Main system power stability	1	lab	Attenuator Network	Resistors	Resistor divider network to scale high voltages down to 3.3V , analog multiplexor for shifting levels	~12	lab
					Analog Multiplexor	HCF4051	For adaptive attenuation	2	120
					Filter Capacitors	Assorted COG/NPO (pF range)	Precision capacitors for the active Butterworth low-pass filter	~8	lab

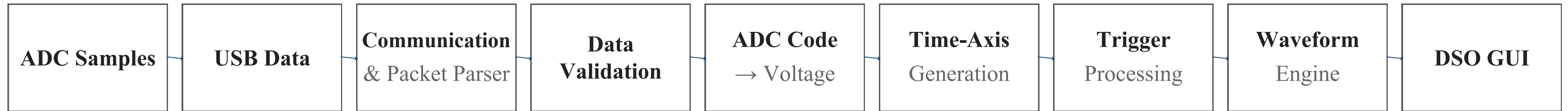
POWER MANAGEMENT				
Component	Recommended Part	Purpose	Qty	Cost
Buck covetrer	Prebuild module	To convert 12v to 5v	1	39
Digital LDO Regulator	AMS1117-3.3 or AP2112K-3.3	Steps down the 5V USB power to 3.3V for the STM32 logic	1	15
Analog LDO Regulator	LP5907-3.3	An ultra-low noise 3.3V regulator <i>specifically</i> for the VDDA (Analog VCC) pin and Op-Amps	1	16
Charge Pump Inverter	TC7660 or ICL7660	Flips +12V to -12V to provide the negative rail for the Op-Amps	1	
Quad Opamp ic	MCP6024	For making buffer stage and some ref volatge	1	190
Filter Inductor (Ferrite)	10µH SMD Ferrite Bead	Isolates the analog ground/power plane from the digital ground	1	lab
Power Caps	10µF Tantalum / Ceramic	Smoothing capacitors for the LDOs and charge pump	4	lab

COMM. AND PROGRAMMING				
USB Connector	USB Type-A Receptacle (16-pin)	Data communication to Host	1	109
USB ESD Protection	USBLC6-2SC6	Protects the STM32's USB pins from static shocks when plugging in	1	9
USB Resistors	22 OHMs Resistor (0603)	USB specification series matching resistors for D+ and D- lines	2	lab
Type-C Resistors	5.1k Omega Resistor (0603)	Placed on CC1 and CC2 lines so a host PC knows to provide 5V power	2	lab
Programming Header	4-Pin Male Header (2.54mm pitch)	SWCLK for the ST-Link V2 programmer	1	
Reset/Boot Buttons	SMD Tactile Push Buttons	Used for hardware reset and Bootloader mode	2	12

Additional Hardware

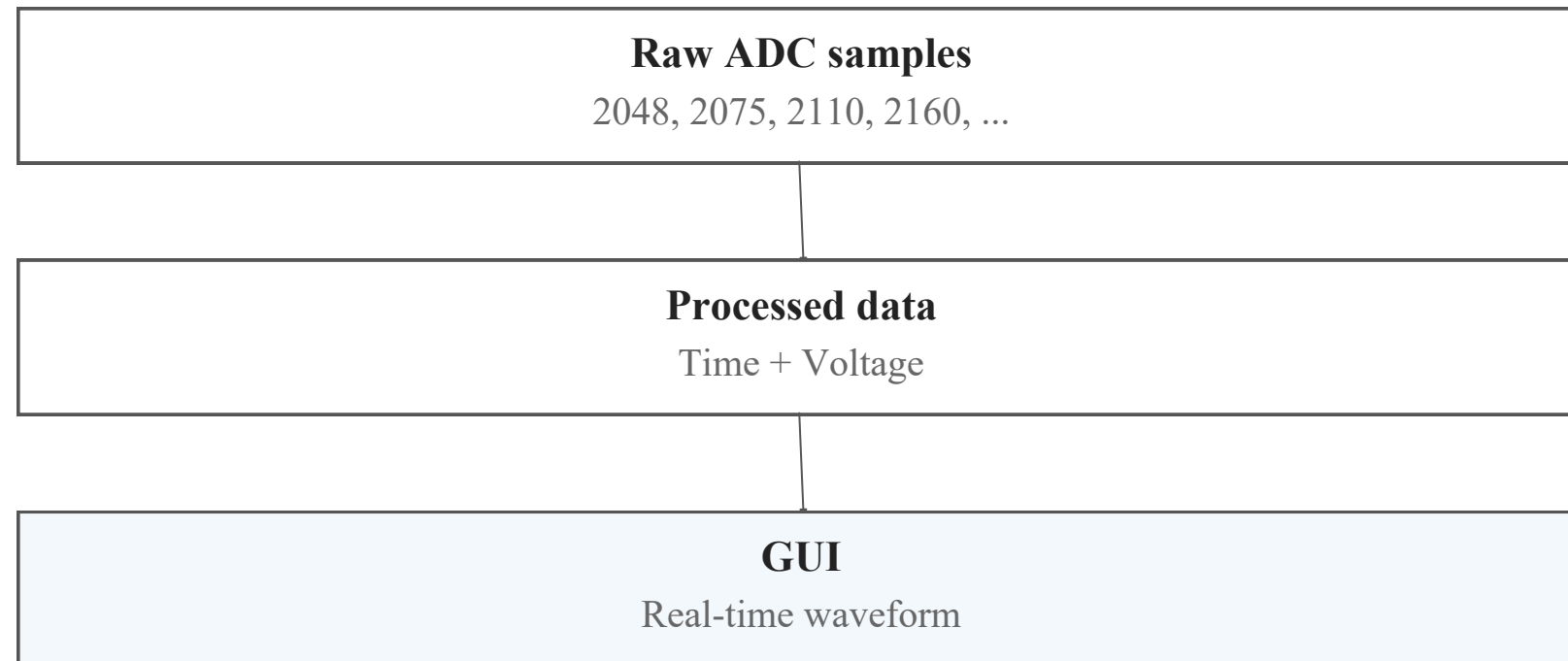
- ST-Link V2 USB Dongle: (Costs ~₹200) To connect this to our 4-pin Programming Header to flash C/C++ firmware onto the STM32.
- Standard P6100 Oscilloscope Probes: Expecting from our lab
- Total estimated budget: ~₹1700

FROM ADC SAMPLES TO REAL-TIME WAVEFORM



The GUI never receives analog voltage — it receives digital ADC samples over USB, which software converts into voltage and time values.

Example



Key processing functions

- Packet reception and buffering
- Packet / header detection
- CRC / data validation
- ADC code to voltage conversion
- Time-axis generation from sampling rate
- Trigger processing for stable display
- Waveform plotting

Sampling rate = 4 MS/s Sampling interval = 250 ns

GUI ARCHITECTURE — PC SOFTWARE PIPELINE

① USB Communication Layer

- Receives the USB 2.0 data stream from the DSO hardware
- Reads incoming packets and manages continuous data reception
- Sends control commands from the GUI back to the DSO when required

② Packet Parser & Data Buffer

- Detects packet header and checks packet length
- Validates CRC / packet integrity
- Extracts ADC samples and stores them in a software buffer for processing

③ Signal Processing Engine

- Converts ADC codes into voltage values using calibration parameters
- Generates the time axis from the sampling rate
- Applies triggering, scaling and optional digital filtering

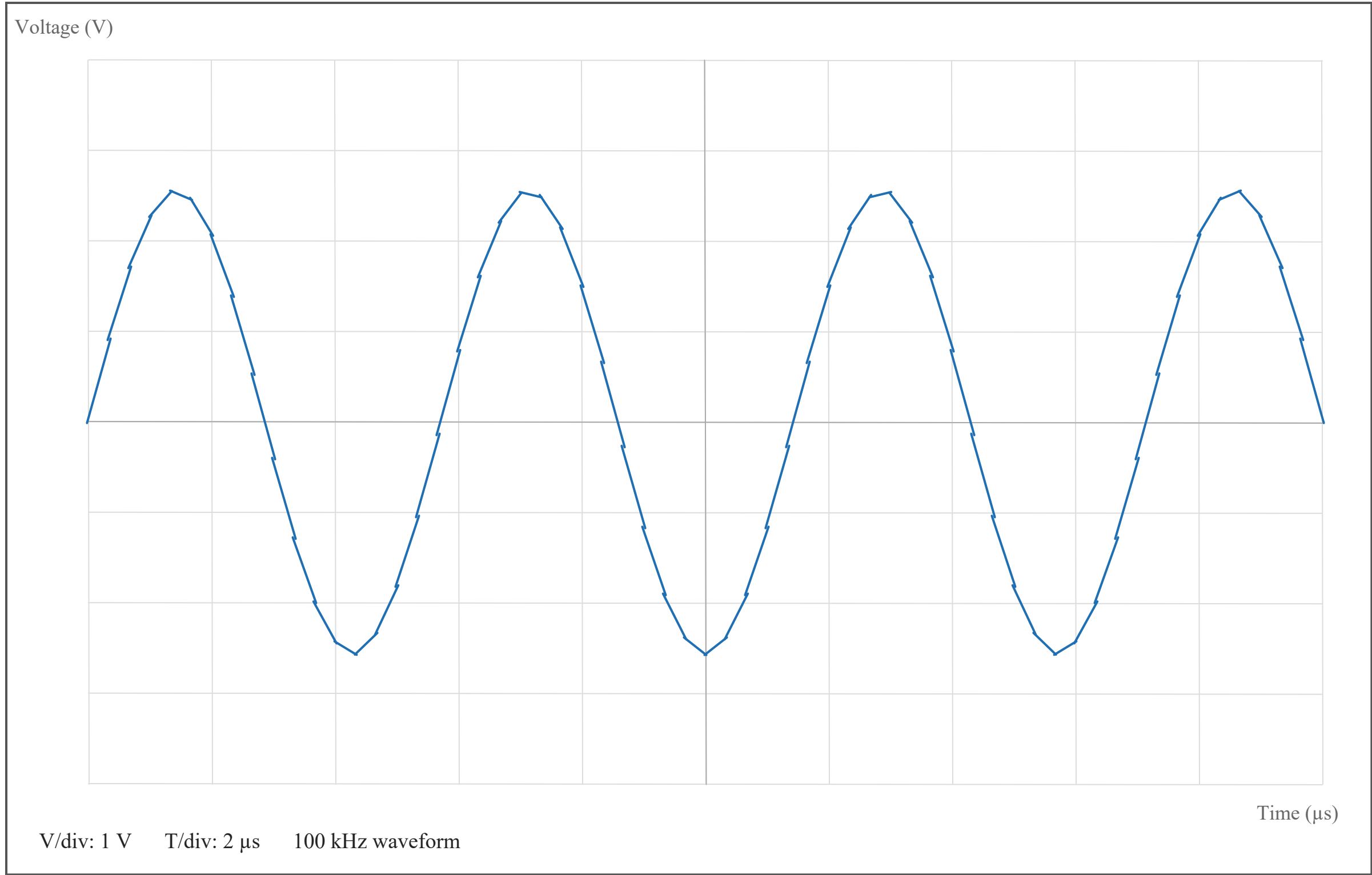
④ Visualization, Measurements & Controls

- Plots the processed samples as a real-time waveform
- Calculates V_{pp} , V_{max} , V_{min} , RMS, frequency and period
- Provides Run / Stop / Single, trigger and scale controls
- User actions are converted into command packets sent back through USB

DSO

Prototype / proposed interface

USB: Connected ✓



Measurements

Vpp	3.30 V
Vmax	1.65 V
Vmin	-1.65 V
RMS	1.17 V
Frequency	100.0 kHz
Period	10.0 μ s

Acquisition control

RUN

STOP

SINGLE

AUTO

Trigger

Source	CH1
Level	0.0 V
Edge	Rising

Scale

Voltage / div	1 V
Time / div	2 μ s

Data

Capture

Save CSV

Clear

The GUI provides real-time waveform visualization, signal measurements, acquisition control, trigger configuration and data export.



THANK YOU!!
ANY QUESTIONS??